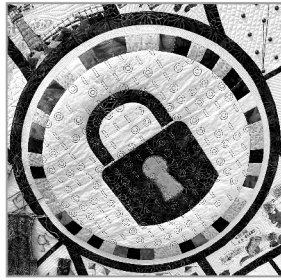


Decoding the Ternary Message in the Cryptography Quilt

The colored ring surrounding the central padlock in the Cryptography Quilt contains a hidden message based on ternary, or base 3, numbers. **Can you decode it?**



The padlock block, with the ternary ring around it.

In our normal base 10, or decimal, number system, we have a ones place, tens place, hundreds place, etc. In ternary, there's still a ones place, but then there's a threes place, a nines place, and a 27s place! Decimal numbers use digits 0 through 9, but ternary only uses 0, 1, and 2.

Example:

$$\begin{array}{ccc} \text{Ternary} & & \text{Decimal} \\ \mathbf{2} & \mathbf{0} & \mathbf{1} \\ \hline (9 & 3 & 1) \end{array} \quad \text{--->} \quad 2 \times 9 + 0 \times 3 + 1 \times 1 = \mathbf{19}$$

Time to make your Decoder Dictionary! Can you finish filling in all the columns? In the quilt the code is written in **three shades of blue** — so instead of coloring the last three boxes, just write the **letter** for each shade:

Color pattern key — one letter per box

L Light blue
= digit **0**

B Blue, the medium shade
= digit **1**

D Dark blue
= digit **2**

Letter	Decimal	Ternary	Color Pattern		
A	1	001	L	L	B
B	2	002			
C	3	010			
D		011			
E					
F		020			
G					
H			L	D	D

Letter	Decimal	Ternary	Color Pattern		
I		100			
J	10				
K					
L					
M			B	B	B
N		112			
O	15				
P					
Q					
R					
S					
T			D	L	D
U					
V					
W					
X					
Y	25				
Z	26	222			

Now go to the quilt and decode the message surrounding the padlock! Start at the top. White squares are spaces between words.

Adapted for black-and-white printing from the Mathemalchemy team's *Decoding the Ternary Message* worksheet — the three shades of blue are written as letters instead of colored in. Mathemalchemy Docent Companion.